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MIL-C-21109C(AS)
20 October 1988
SUPERSEDING
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10 August 1966

MILITARY SPECIFICATION

CLOTH, COATED, NYLON, RAFT BOTTOM

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for nylon base natural or synthetic natural rubber coated cloth.

1.2 Classification The raft bottom coated cloth shall be of the following types as specified (see 6.2).

Type I	-	7 oz
Type II	-	14 oz.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

*	PPP-P-1136	Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

Military

	MIL-C-21108	Cloth, Nylon, Raft Bottom

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Federal

FED-STD-141	Paint, Varnish, Lacquer, and Related Materials, Methods of Inspection, Sampling, and Testing
FED-STD-191	Textile Test Methods
FED-STD-595	Colors
FED-STD-601	Rubber, Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

* (Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN:NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.3), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Materials. The coated cloth shall consist of nylon base cloth uniformly coated with a natural rubber or synthetic natural rubber base compound.

3.2.1 Base cloth. The base cloth shall conform to the requirements of MIL-C-21108, Type I or II as required in 3.3.

3.2.2 Coating. The compound used in coating the base cloths shall contain not less than 80 percent by volume of new plantation natural rubber or synthetic natural rubber. The compound shall also contain softeners, curing agents, antioxidants and reinforcing materials. The compound used in the outer coatings shall be pigmented during the compounding process so that the cured coating shall have a uniform and permanent color as specified in 3.5. The pigment shall be of such fineness that 100 percent of the pigment shall pass through a standard 325 mesh screen. The foundation coating and rubber coating compounds shall not be injurious to the base cloth and shall contain no compounding ingredients which would bloom to the surface or which would adversely affect the specified properties of the finished coated cloth. The compounding ingredients shall be prepared in such proportions as to cure properly and provide proofing films which shall be insoluble in water to the

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extent that the coating compound water extractable matter shall amount to no more than 1 percent by weight (see 4.6.6). The coating compound, cured in sheet form to the same degree of cure and in the same manner as the finished product, shall conform to the physical requirements specified in Table I.

TABLE I. Physical properties of cured coating compound.

Characteristic	Requirement	FED-STD-601 Test Method
Tensile strength		4111
Original, psi	2300 min	--
After heat aging for 168 hours at 100 ± 1 1°C (212 ± 2°F), percent loss	30 max	7221
Elongation		4111
Original, percent	450 min	--
After heat aging for 168 hours at 100 ± 1.1°C (212 ± 2°F), percent loss	15 max	7221
Shore hardness		
Original	50 ± 10	3021
After heat aging for 168 hours at 100 ± 1.1°C (212 ± 2°F), change	±5	

3.2.2 1 Coating adhesion. The coating adhesion test specimens described in 4.4.1.1 shall have a coating adhesion of not less than 7 0 pounds per inch of width when tested as specified in 4.6 6.4.

3.3 Construction and weight distribution. The construction and weight distribution of components of the finished cloth shall be as specified in Table II

* TABLE II. Construction and weight distribution

Coated Cloth	Rubber Coating Compound Side One	Nylon Cloth MIL-C-21108		Rubber Coating Compound Side Two	Total Weight Ounces Per Square Yard
		Type I	Type II		
Type I	2 ± 0.2	2.7 max	---	2 ± 0.2	6.3 to 7.1
Type II	4.2 ± 0.2	---	5.5 max	4.2 ± 0.2	13.3 to 14.3

3.3.1 Manufacture. A foundation coating, which shall be compatible with the base cloth and the remainder of the coating compound, shall be applied to the base cloth in order to achieve the required adhesion The coating compound shall then be applied to the base cloth by means of a spread coating operation. The rubber coating compound shall be applied in a sequence of

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operations so that both sides of the cloth shall be coated and built up to the construction shown in Table II. The pigment shall be incorporated in the outermost coatings. The coated cloth shall be cured to meet the requirements of this specification. The cured coated cloth may be lightly dusted with talc or zinc stearate.

3.4 Age. Cloth offered for acceptance under this specification shall be from current production, and not more than 90 days old prior to release for shipment. The date of completion of the coating operations shall be stamped at the end of each roll and on the identification tags. The ink used for stamping this date of manufacture on the roll shall be free of manganese and copper (see 4.6.4 and Table VI).

3.5 Color. Unless otherwise specified, the uniform permanent color of the finished coated cloth shall be an approximate match to FED-STD-595 color number 12197 (see 4.6.8).

3.6 Properties of coated cloth. The coated cloth properties shall conform to the requirements specified in Table III and in the following paragraphs when tested as indicated.

TABLE III. Properties of coated cloth.

Characteristic	FED-STD-191 Test Method	Requirement	
		Type I	Type II
Breaking strength, grab method	5100		
As received, pounds, min (see 4.5.9)			
Warp		150	300
Filling		140	300
After heat aging for 168 hours at 93 ± 1.1°C (200 ± 2°F)	5850		
breaking strength loss, percent, max		10	10
Hydrostatic resistance, pounds per sq. in., min (see 4.5.9)	5512		
As received		50	100
After heat aging for 168 hours at 93 ± 1.1°C (200 ± 2°F)	5850	50	100
Blocking at 93 ± 1.1°C (200 ± 2°F), rating, max (see 4.5.11)	5872	2	2

3.6.1 Resistance to heat aging. The coated cloth shall not become stiff and brittle or soft and tacky after heat aging for 168 hours at 93 ± 1.1°C (200 ± 2°F) (see 4.6.9).

3.6.2 Resistance to low temperature exposure. The coated cloth shall show no signs of cracking after low temperature exposure for 4 hours at -54 ± 1.1°C (-65 ± 2°F) and testing as specified in 4.6.10.

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3.7 Dimensions. Unless otherwise specified, the coated cloth shall be furnished in rolls with a minimum length of 250 yards and a maximum length of 350 yards The minimum length of pieces shall be 50 yards Width shall be as specified by the procuring activity.

3.8 Workmanship The finished coated cloth shall be made from clean, evenly woven base cloth evenly coated and uniformly processed to produce the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels specified herein.

4. QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4 1.1 Responsibility for compliance All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.2 Classification of inspections The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

* 4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.

4.4 First article inspection The first article inspection of the raft bottom coated cloth shall consist of all the examination and test requirements of this specification.

* 4 4.1 First article inspection samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, the contractor shall furnish first article samples of each type of cloth for inspection to determine conformance with this specification. The first article inspection

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sample shall consist of required specimens of coated cloth, base cloth, a coating adhesion test specimen prepared as described in 4.4.1.1, and sheets of cured coating compound. The size and quantities of components of the first article sample shall be specified by the procuring activity. When a contractor is in continuous production of the cloth from contract to contract, submission of further first article inspection samples, on the new contract, may be waived at the discretion of the procuring activity. Approval of the first article inspection samples or the waiving of the first article inspection does not waive the requirements of performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2.1). The first article inspection sample shall be identified by securely attached tags or labels durably marked with the following information:

Sample for First Article Inspection
CLOTH, COATED, NYLON, RAFT BOTTOM
Name of manufacturer
Name of distributor (if applicable)
Product designation (manufacturer's)
Date of manufacture
Submitted by (name) (date) for first article inspection in accordance with the requirements of MIL-C-21109C(AS)

4.4.1.1 Coating adhesion test specimen. The coating adhesion test specimen shall be prepared by the manufacturer (coater). Each of 2 pieces of base cloth (see 3.2.1) measuring 12 by 12 inches shall be coated with the uncured coating compound of 3.2.2. The specimen shall be constructed so that adhesion will be measured in the warp direction. The 2 pieces shall be placed together, coating to coating, except for a 1 inch wide separation along one edge allowed by insertion of paper. The specimen shall be cured in the same manner and degree of cure as the finished product. Specimens shall be made to represent the construction and weight distribution as shown in Table II.

* 4.4.2 Test report. Two copies of the contractor's test report, containing complete test data for the end item and for each component required herein and referring specifically to the applicable paragraphs, shall be submitted with the first article inspection sample.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of all the examination and test requirement specified herein except for coating adhesion (4.6.6.4), resistance to heat aging (4.6.9), and resistance to low temperature exposure (4.6.10). All the sampling and inspection levels shall conform to MIL-STD-105.

4.5.1 Quality conformance inspection samples.

4.5.1.1 Inspection lot. A lot of coated cloth shall consist of a quantity of the product manufactured from 10 batches of coating compound or fraction thereof, manufactured under essentially the same conditions, and offered for delivery at one time. The lot size shall be expressed in units of 1 yard each, unless otherwise specified.

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4.5.1.2 Sample for yard-by-yard examination. A sample for the yard-by-yard examination shall be selected in accordance with MIL-STD-105, Inspection Level II. The acceptable quality level shall be 2.5 percent for major defects, and 10.0 percent for overall defects (major and minor combined) per 100 units (yards). Sufficient rolls shall be selected at random from the lot so that the required sample yardage shall be obtained by examining approximately 25 consecutive yards out of each sample roll (see 4.6.2). The sample for yard-by-yard examination of the base cloth shall be selected in accordance with MIL-C-21108.

4.5.1.3 Sample for overall, dimension and age examinations. A sample for the overall examination (see 4.6.3) and the dimensions and age examinations (see 4.6.4) shall be selected at random in accordance with Table IV. The unit of product for these examinations shall be one piece.

TABLE IV. Sample size.

Lot Size in Yards	Sample Size in Pieces	Max Number of Defects Acceptable in Sample
Up to 1300 1/	3	0
1301 to 3200	5	0
3201 to 8000	7	0
8001 to 22,000	10	0
22,001 to 110,000	15	1
110,001 and over	25	1

1/ If a lot contains fewer than three (3) pieces, each piece in the lot shall be examined.

4.5.1 4 Sample for quality conformance inspection tests. The sample unit for purposes of quality conformance inspection tests shall be 2 yards. The number of sample units to be tested shall be in accordance with Inspection Level S-1 of MIL-STD-105, considering the lot size in units of 2 yards. The acceptable quality level shall be 4.0 percent defective Random sampling shall be accomplished so that each 2-yard sample shall be taken from different rolls. Each sample shall be suitably tagged to identify it with the roll sampled. The test samples shall include four 6 by 6 by 0.07 to 0.08 inch sheets of cured coating compound from each batch and the coating compound samples shall be identified with the lot number. The test samples of coating compound and coated cloth shall have been cured to the same degree of cure. The sample for tests of the base cloths shall be selected in accordance with MIL-C-21108.

4 5.1.5 Sample for examination of preparation for delivery requirements. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The sample size shall be determined in accordance with MIL-STD-105, Inspection Level S-2. The acceptable quality level shall be 4.0 percent defective.

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4.6 Inspection methods. Unless otherwise specified, inspection shall be in accordance with the provisions of MIL-STD-105 and testing shall be in accordance with the conditions and methods of FED-STD-191. Unless otherwise specified, the physical and chemical values specified in Section 3 apply to the average of the determinations made on the unit of product

4.6.1 Preparation for delivery examination. Each sample shipping container fully prepared for delivery shall be examined to determine compliance with the preservation, packing and marking requirements of this specification. Examination shall be made for the following defects:

Markings, exterior and interior	Missing; incorrect; illegible; of improper size, location, or method of application
Materials	Component materials nonconforming, missing, damaged, or otherwise defective affecting serviceability
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, etc. Bulging or distortion of containers.
Weight	Gross weight exceeds requirement.

4.6.2 Yard-by-yard examination. The required yardage of each piece of cloth in the sample (4.5.1.2) shall be examined over a bank of lights for defects as listed in MIL-C-21108 for the base cloth and as listed in Table V for the coated cloth. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only one defect shall be counted A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs.

TABLE V. Classification of defects for coated cloth.

Defects	Classification	
	Major	Minor
Blisters or lumps		
1/4 square inch or less in combined directions, clearly visible 1/		X
Over 1/4 square inch in combined directions, clearly visible 1/	X	
Coating faults		
Blooming	X	
Creases	X	
Folds - winding on tubes		X
Uneven coating		X
Heavily coated areas, clearly visible 1/		X
Thin areas where coating compound is insufficient	X	

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TABLE V. Classification of defects for coated cloth (continued)

Defects	Classification	
	Major	Minor
Coating missing	X	
Cuts	X	
Foreign matter incorporated in coating	X	
Pinholes	X	
Selvage defects affecting coating adhesion and resulting in uncoated area	X	
Splice mark clearly visible on coated side 1/		
If coating is not removed		X
If coating is missing	X	
Stains in coating surface		
Any spot or stain (compound, oil, dirt) over 1 inch in length in any direction, clearly visible 1/		X
Streaks		
Shade streaks in coating, clearly visible 1/	X	
Tears	X	

1/ At normal examination distance (approximately 3 feet).

4.6.3 Overall examination. The required number of pieces of coated cloth in the sample (4.5.1.3) shall be examined for defects as listed in Table VI. Each defect shall be counted no more than once in each piece examined.

TABLE VI. Overall defects

Selvage curled, folded, rolled or slack continuous or intermittently throughout the piece
Overall uncleanness or objectionable odor.
Uneven coating throughout the piece
Off-shade, shaded end to end, side to side, side to center or throughout the piece.
Mottled, cloudy or streaky throughout the piece.
Date of coating application missing from end of roll or identification tag.

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4.6.4 Dimension and age examination. Each piece of coated cloth in each roll in the sample (4.5.1.3) shall be counted and measured, and the age of each roll shall be determined. Width and length shall be measured as specified in Methods 5020 and 5010 of FED-STD-191. The maximum number of defects allowable is shown in Table IV. The defects are as follows:

Any gross length or width of a roll less than minimum specified.

Any roll containing more than the maximum number of pieces per roll as specified.

Any gross length of a piece less than the minimum length of the shortest piece in a roll as specified.

Any gross length less than the gross length marked on the piece ticket by 2 yards or more.

The lot shall be considered unacceptable if any of these defects are present.

Total of the actual gross lengths of pieces in the sample is less than the total of the gross lengths marked on the piece ticket

Age of any sample roll exceeds the limit specified (3.4).

Width of the material is not within tolerances

The absence of manganese and copper in the ink used for stamping the date of manufacture on the roll shall be verified (3.4).

4.6.5 Base cloth. The base cloths shall be examined and tested to determine conformance to MIL-C-21108 and the requirements of this specification (3.2.1).

4.6.6 Coating. Conformance to the coating compound ingredient requirements shall be determined by appropriate examination and testing in accordance with 3.2.2 and by acceptance of a certificate of compliance.

* 4.6.6.1 Solubility. Solubility in water shall be determined in accordance with Method 6631 of FED-STD-601.

* 4.6.6.2 Hardness. Shore durometer hardness (after 15 seconds) shall be determined in accordance with Method 3021 of FED-STD-601.

4.6.6.3 Tensile strength and elongation. The tensile strength and elongation of cured coating compound samples shall be determined after exposure to heat aging as well as on the original samples. The determinations of tensile strength and elongation shall be made in accordance with Method 4111 of FED-STD-601. The Type III die shall be used to prepare specimens. The specimens to be heat aged shall be exposed for 168 hours in an oven maintained at a temperature of 100 ± 1.1°C (212 ± 2°F) in accordance with Method 7221 of FED-STD-601 (see Table I).

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4.6.6.4 Coating adhesion The coating adhesion test shall be performed in accordance with Method 8011 of FED-STD-601 Adhesion shall be measured in the warp direction (3.2.2.1).

4 6 7 Construction, weight distribution and manufacture. Conformance to the construction and weight distribution requirements of Table II and manufacture requirements of 3.3.1 shall be determined by examination in accordance with Method 5041 of FED-STD-191.

4.6.8 Color The color of the coated cloth shall be compared with the standard color chip specified in 3 5 in accordance with Method 4250 of FED-STD-141.

4.6.9 Resistance to heat aging. Resistance to heat aging of the coated cloth shall be determined in accordance with 5850 of FED-STD-191. The test specimens shall be heated in the oven for 168 hours at a temperature of 93 ± 1°C (200 ± 2°F) After the heat aging, the specimens shall be examined for stiffness, brittleness, softness and tackiness (3.6.1). Heat aged specimens shall be cooled and conditioned under standard atmospheric conditions for 4 hours, and then separate specimens shall be tested as indicated in Table III for breaking strength by the grab method in accordance with Method 5100 of FED-STD-191, and hydrostatic resistance in accordance with Method 5512 of FED-STD-191 These properties of the exposed cloth shall be compared with those of the cloth in the as received condition

4 6 10 Resistance to low temperature exposure. Specimens of coated cloth shall be separately folded into quarters, the folds being made along the warp and filling directions of the base cloth (see Table II) The folds shall be sharply creased and each folded specimen shall be placed on a polished metal plate and completely covered with another metal plate. A weight shall be centrally placed on each covering plate The covering plate and weight shall weigh a total of 5 pounds These assemblies shall be subjected to a temperature of -54 ± 1.1°C (-65 ± 2°F) for a total of 4 hours. After 3-1/2 hours of exposure to this temperature, the specimens shall be unfolded carefully and refolded so that the folds are reversed. The unfolding and refolding shall be performed in the cold box at the specified low temperature After refolding, and with the covering plate and weight replaced, the assembly shall remain in the cold box at the specified low temperature, for the remainder of the 4-hour period. The specimens shall then be removed, conditioned under standard atmospheric conditions for 2 hours, and examined for evidence of cracking (see 3.6.2).

4.6.11 Blocking. The blocking test rating of the coated cloth shall be determined in accordance with Method 5872 of FED-STD-191 except the exposure temperature shall be 93 ± 1.1°C (200 ± 2°F) (see Table III).

5. PACKAGING

5.1 Preservation, packing and marking. Cloth shall be preserved, packed and marked in accordance with PPP-P-1136, except that, prior to rolling, the coated face of the fabric shall be dusted with a suitable talc to prevent

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tackiness. The levels of preservation and packing shall be as specified by the procuring activity (see 6.2). In addition to the markings required, each tag shall be marked with the following information: "Date of Application of Coating." In addition, the following precautionary marking shall appear on all containers:

"STORE IN A COOL DRY PLACE"

6. NOTES

6.1 Intended use. The coated cloths covered by this specification are intended for use in the manufacture of life raft bottoms. Type I is for one-man rafts, cylinder carriers, oral inflation valve pockets, sea anchor mooring patches, and accessory patches used in the manufacture of pararafts and packet rafts. Type II is for the multi-place life raft bottoms.

6 2 Ordering data. Procurement documents should specify the following·

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Type required (see 1.2).

d. Color required if other than as specified in 3.5

e. Width and width tolerance required.

f. Dimensions and roll put-up requirements, if other than as specified in 3.7.

g. Quantity required.

h. Selection of applicable levels of packaging and packing (see 5.1).

6.3 First article. When a first article inspection is required, the item should be a first production item as required by 4 4.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

* 6 4 Subject term (key word) listing.

Cloth
Coated
First article
Natural rubber
Nylon
Raft bottom
Synthetic natural rubber

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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